Pre-Calculus II — Assignment 1

Department of Information Technology | BSIT - Semester VI

Topics Covered: Complex Numbers, Arithmetic Series & Geometric Series

Total Marks: 25

Course Instructor: Faculty of IT

Submission Deadline: Strict / As Announced

General Instructions: Solve all four questions with rigorous mathematical steps. Neatness, proper algebraic justification, and conceptual clarity will be evaluated. Calculators are allowed only for verification. Note that final answers without complete step-by-step working will not be accepted.

Assignment Questions

Question 1: The Complex Roots Puzzle

[6 Marks]

Let z be a complex number satisfying the equation $z^3 + 8i = 0$.

1. Find all distinct roots of this equation in polar form $r(\cos\theta + i\sin\theta)$ and Cartesian form $a + bi$.
2. Plot these three roots on a complex plane (Argand diagram conceptually) and prove that the sum of these three roots equals zero:

$$z_1 + z_2 + z_3 = 0$$

Question 2: Arithmetic Series & Sequence Interlock

[6 Marks]

The sum of the first n terms of a series is given by the quadratic formula $S_n = 3n^2 + 5n$.

1. Find the n-th term (a_n) of this sequence and prove that the sequence is an Arithmetic Progression (A.P.). State its first term and common difference d.
2. If the k-th term of this sequence is equal to 164, determine the exact value of k.
3. Find the sum of all terms from 10^{th} to 20^{th} terms inclusive.

Question 3: Geometric Series Convergence & Infinite Sum Trap

[7 Marks]

Consider an infinite geometric series whose sum to infinity is $S_\infty = 4$, and the second term of the series is $a_2 = 3/4$.

1. Set up the equations in terms of the first term a and common ratio r, and show that r satisfies a quadratic equation. Solve for all possible values of r and the corresponding first term a.
2. For the valid value of r, calculate the sum of the first 6 terms (S_6) and find the percentage error when approximating S_∞ by S_6:

$$\text{Percentage Error} = |(S_\infty - S_6) / S_\infty| \times 100\%$$

Question 4: The Grand Synthesis (Complex Numbers meets Series)

[6 Marks]

Let ω be a complex cube root of unity (i.e., $\omega^3 = 1$ and $1 + \omega + \omega^2 = 0$, where $\omega \neq 1$).

1. Evaluate the finite geometric series sum involving powers of ω:

$$S = 1 + 2\omega + 3\omega^2 + 4\omega^3 + 5\omega^4 + 6\omega^5$$

   (Hint: Simplify each term using the periodic properties of powers of ω before adding, or treat it as an Arithmetico-Geometric Series).

2. Using your result, express the final simplified complex number in standard form $a + bi$ and compute its modulus $|S|$ and argument $\arg(S)$.